

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SCSR2213 NETWORK COMMUNICATIONS

Task#4

Making The Connections - LAN And WAN

Group 4: Linux

SECTION: 5

Lecturer: Dr. Adnan Khalid

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In the earlier tasks, we laid a strong foundation for the project. In Task 1, our team, "Linux," was formed, and we developed a comprehensive floor plan for the new two-story Faculty of Computing building, along with documenting our preliminary discussions in meeting minutes.

Task 2 involved a thorough analysis where we formulated over 10 focused questions to better understand the requirements and gather insights. Through detailed research and consultations, we assessed the current and future needs of the project, evaluated its feasibility, and made necessary revisions to our design to align with the expectations of stakeholders.

For Task 3, we selected the most suitable LAN devices that meet the objectives of the case study, ensuring that the building would be equipped with a robust and secure network infrastructure capable of meeting both current and future demands.

In this task, we will focus on designing the physical network infrastructure, including strategic placement of devices, cabling, and inter-floor connections to seamlessly integrate the new rooms into the overall network design. We will carefully plan for both horizontal and vertical cabling, choosing the right media (twisted pair, fiber optic, or wireless) based on the specific requirements. The ultimate goal is to create a reliable, high-performance network infrastructure for the new building.

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- **Working areas: First Floor Working Areas**

The ground floor of the building, as shown in Figure 1, includes six distinct work areas.

Work Area 1 is the Server Room, which is essential for the IT infrastructure. It houses five server racks responsible for managing data, applications, and network performance throughout the building. In addition, there are two workstations in the room, allowing IT staff to monitor and manage server operations effectively. These devices are critical to ensuring the smooth operation of the facility and supporting the overall network.

Work Area 2 is the Hybrid Classroom, which is equipped with two large TVs for displaying teaching materials and presentations, making it easier for instructors to share content with students. The classroom also contains a workstation for the instructor to facilitate lesson delivery and manage activities, ensuring a better learning experience.

Work Area 3 is the Video Conference Room, dedicated to virtual meetings and collaborations. This room features a large TV for video calls, presentations, and screen sharing, enabling participants to interact seamlessly.

Work Area 4 is the Lecturer's Office, consisting of 18 individual rooms. Each room includes a workstation, providing lecturers with a private, quiet space to carry out administrative and academic work.



Figure 1: Work Areas in ground floor

Work Area 5 is the Academic Office, a space designed for both individual and collaborative work. It is equipped with 27 workstations to accommodate staff and students. Additionally, there are three TVs for presentations and information display. To support document management, two printers and two photocopiers are available.

Work Area 6 is the Meeting Room, which has a large TV for content display, facilitating meetings and enabling participants to share materials and collaborate effectively.

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This setup ensures that each work area is optimally designed to meet the needs of different functions within the building.

Figure 2 illustrates the work areas located on the first floor, which consists of five main workspaces.

Work Area 7 is the Student Lounge, designed as a space for students to relax, socialize, and collaborate. It features comfortable sofas, a large meeting table surrounded by chairs, and small tables, offering a flexible environment suitable for both individual and group activities. The lounge is also equipped with a TV for informational purposes and a photocopier to assist students with printing and copying tasks.

Work Area 8 is the Cisco Network Lab, a specialized facility for networking studies and hands-on applications. This lab is furnished with 31 workstations, including 30 for students and 1 for the lecturer. Additionally, there is a small storage room with three cabinets to organize tools, equipment, and other materials essential for networking activities, ensuring the lab remains organized and functional.

Work Areas 9 and 11 are the General Labs, each of which is equipped with 31 workstations: 30 for students and 1 for the lecturer. Both labs also feature a small storage area with three cabinets to store tools, equipment, and materials needed for various lab activities.

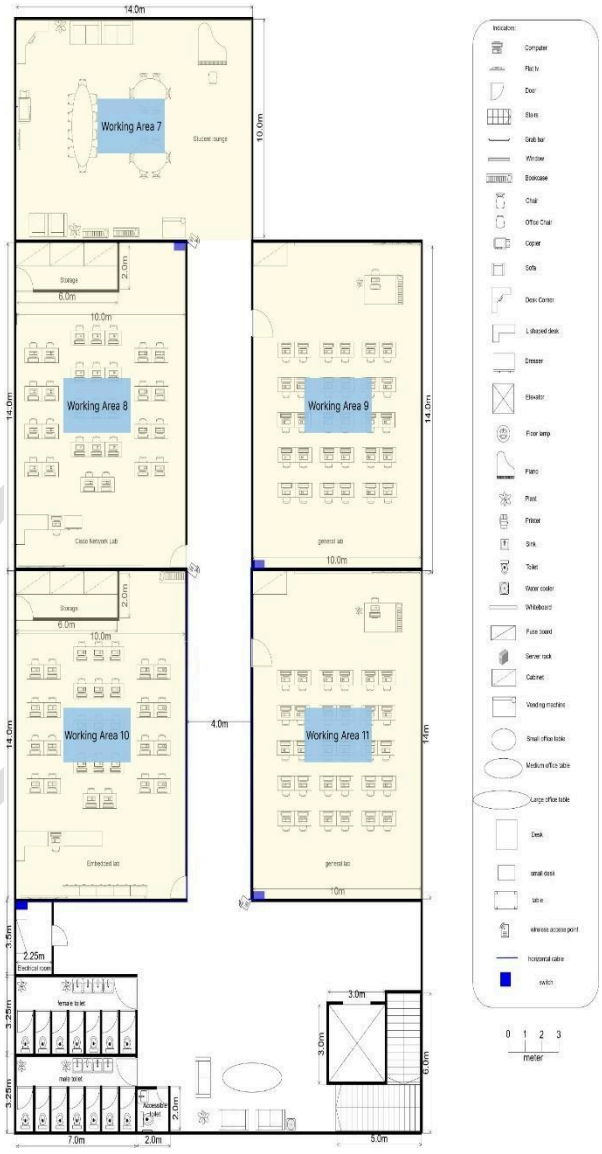


Figure 2: Work Areas in first floor

Work Area 10 is the Embedded Lab, dedicated to hardware programming and embedded systems projects. It includes 31 workstations, 30 for students and 1 for the lecturer. Like the other labs, the Embedded Lab also has a small storage space with three cabinets to store tools, components, and other necessary resources.

In summary, Figures 4.1.1 and 4.1.2 showcase a total of 11 work areas across the building, each carefully designed to support the diverse functions and activities within the facility.

The distribution and connection of devices

3.1 Network Distribution Diagram:

This diagram illustrates the network connections between the devices in the faculty building. It mainly highlights the wired and wireless links that connect the server room to the other rooms across the building.

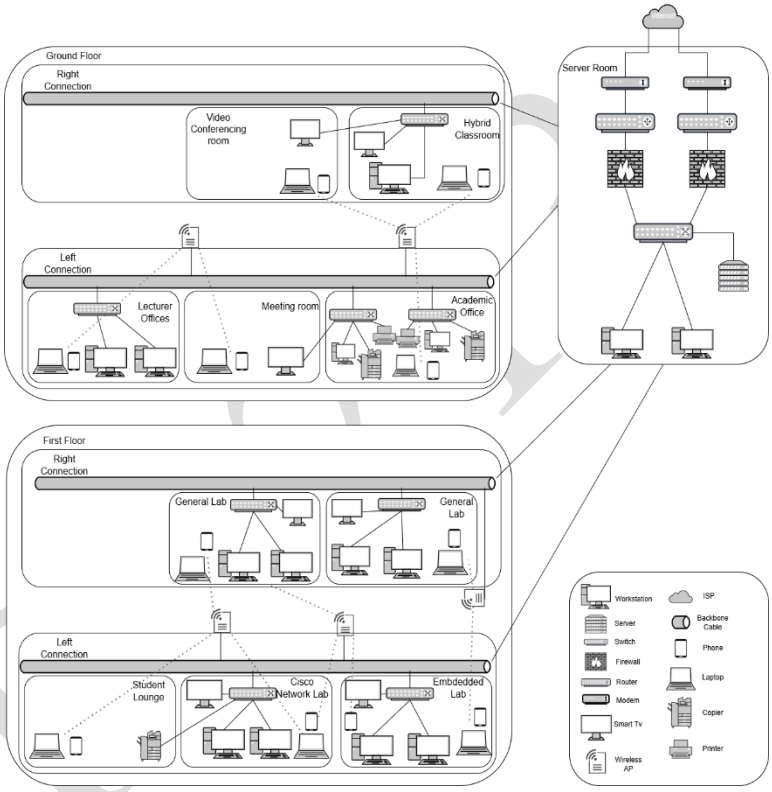


Figure 3.1.1 Network Distribution Diagram

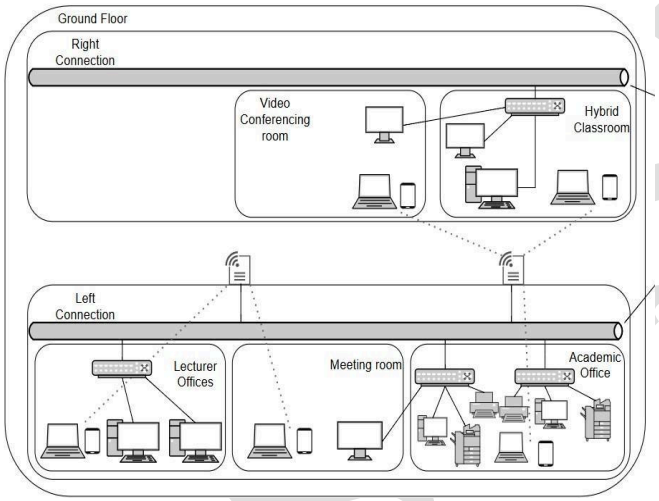


Figure 3.1.1.1 Network Distribution Diagram - Ground Floor

Ground Floor:

As shown in Figure 3.1.1.1, there are two primary connections from the server room: one on the right and one on the left. The right connection includes a single switch located in the Hybrid Classroom, which then extends to the Video Conference Room. On the left, there are three switches: two in the Academic Office to support the heavy workload, and one in the Electrical Room that connects to the lecturers' offices. Additionally, the ground floor is equipped with two wireless access points to ensure wireless internet access throughout the area.

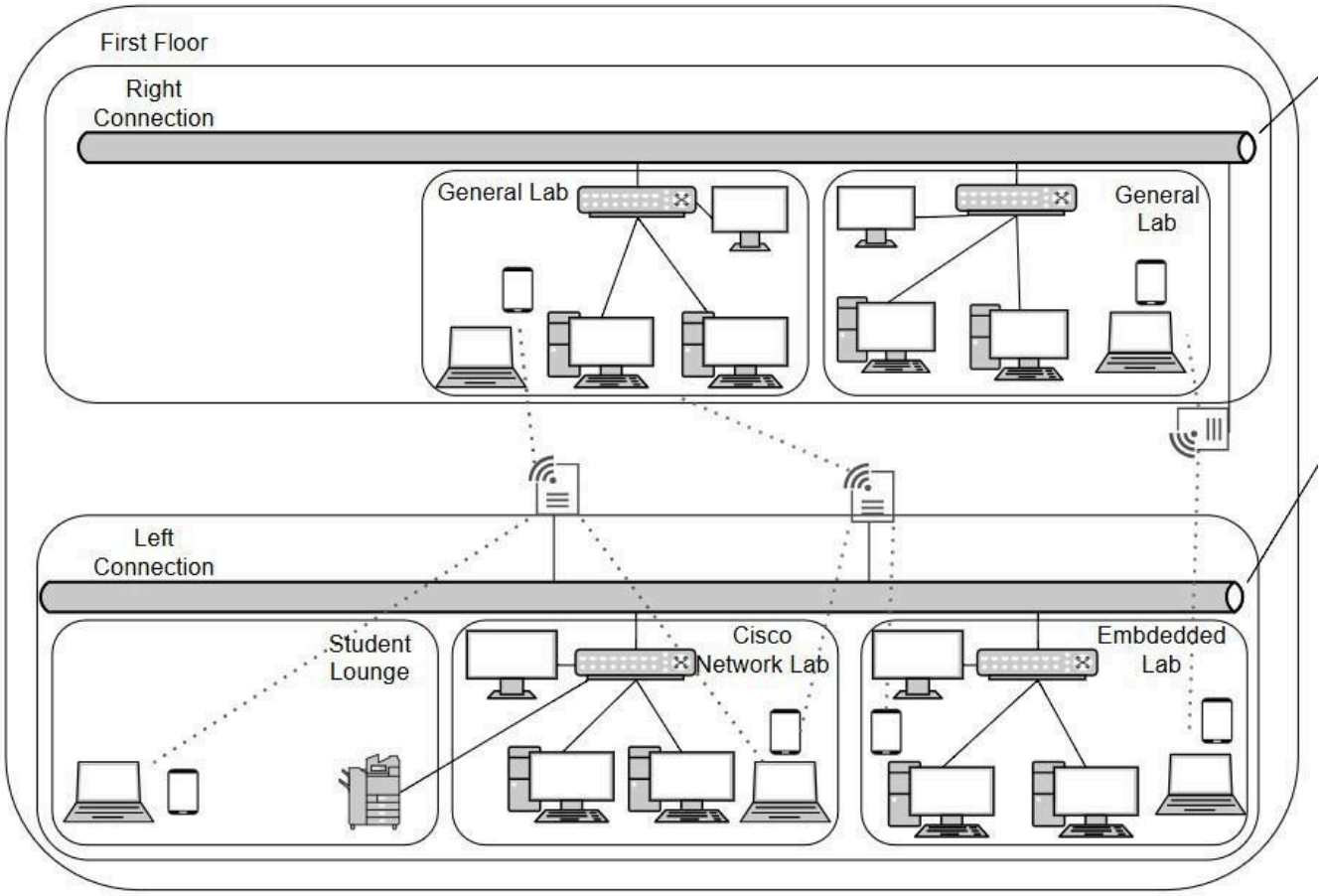


Figure 3.1.1.2 Network Distribution Diagram - First Floor

First Floor:

The first floor follows a similar setup to the ground floor, as shown in Figure 3.1.1.2, with both right and left connections. The right connection is linked to two switches, one in each General Lab, and includes a wireless access point. On the left side, there are also two switches: one in the Embedded Lab and one in the Cisco Network Lab. Additionally, the first floor is equipped with two wireless access points to ensure complete wireless coverage throughout the area.

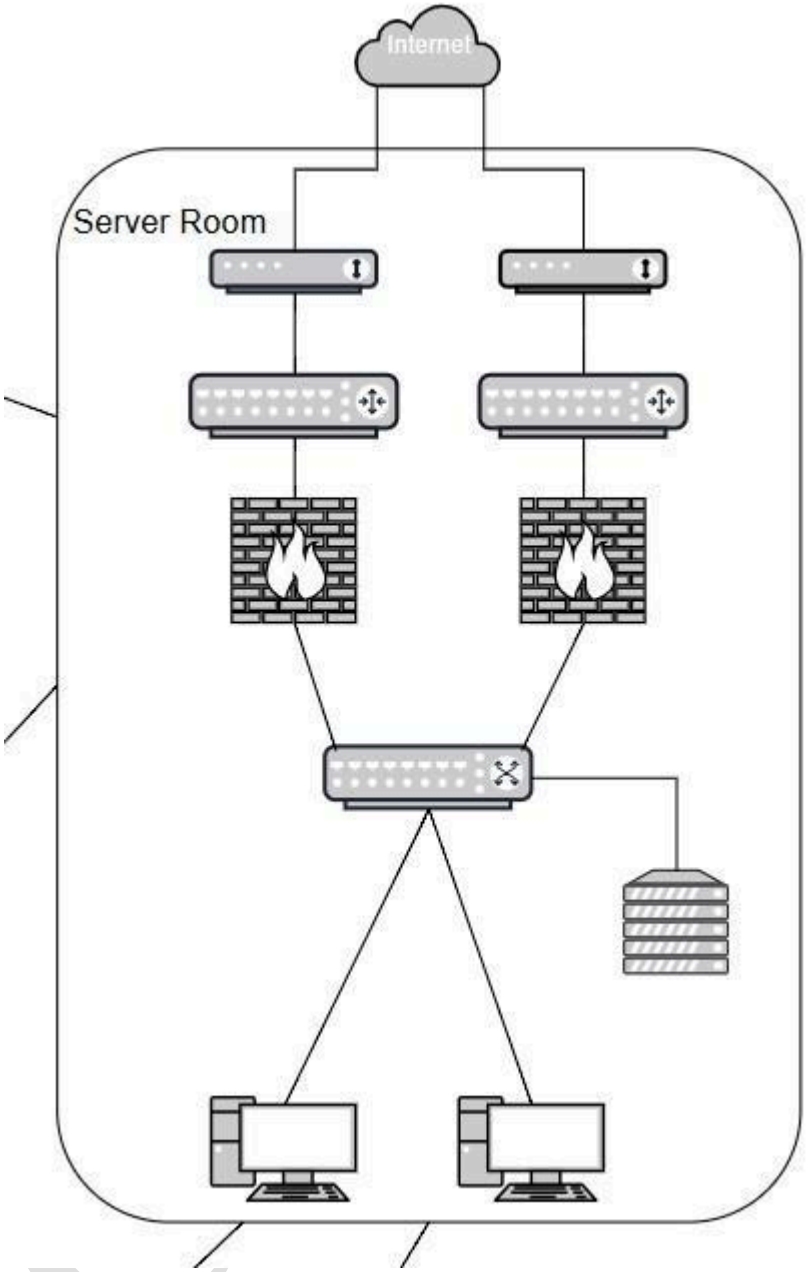


Figure 3.1.1.3 Network Distribution Diagram - Server Room

Server Room:

As seen in Figure 3.1.1.3, the server room utilizes a 2-tier network architecture, which allows it to manage the network traffic of the faculty building efficiently. This architecture is also more cost-effective compared to a 3-tier setup. The server room is equipped with an additional modem, router, and firewall to ensure redundancy, making the network more resilient. If one router fails, the other can continue managing the traffic until the problem is resolved. Furthermore, the server room contains two workstations that are directly connected to the core switch, allowing for better control over the network. The server itself is also linked to the core switch to ensure a stable and reliable connection.

- **Room Level Network Distribution Diagram:**

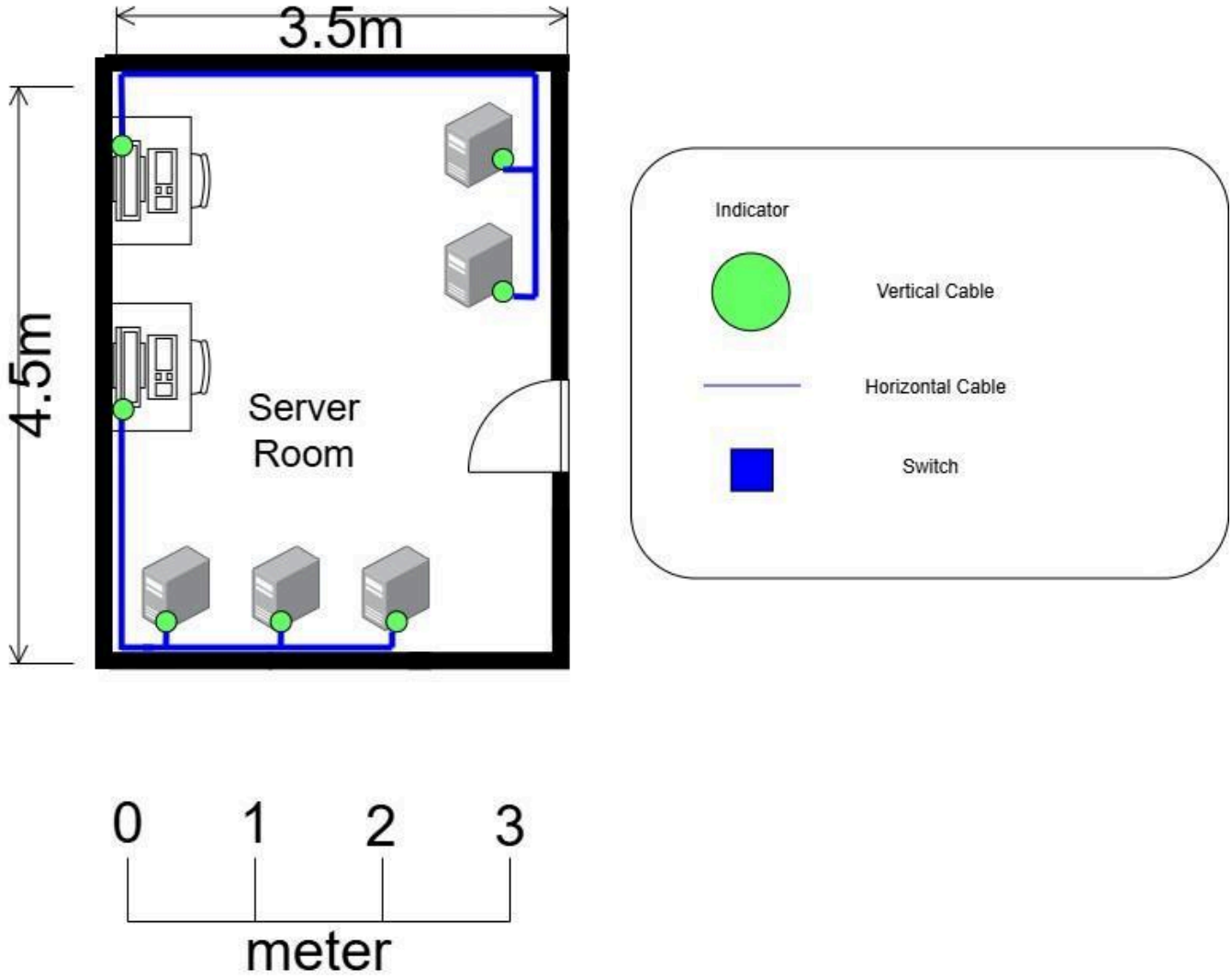


Figure 3.2.1 : Server room

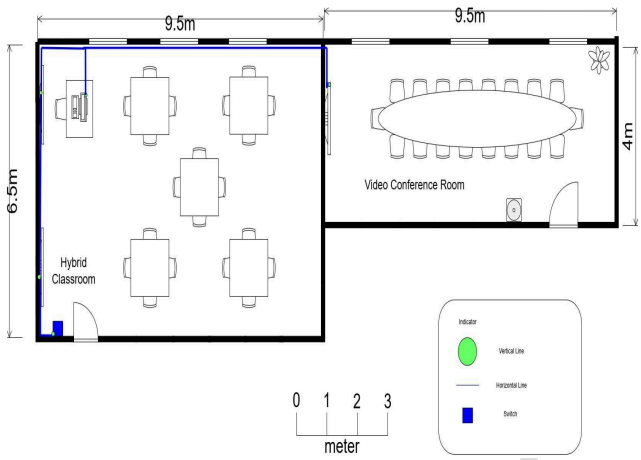


Figure 3.2.2 : Hybrid Classroom & video conference room

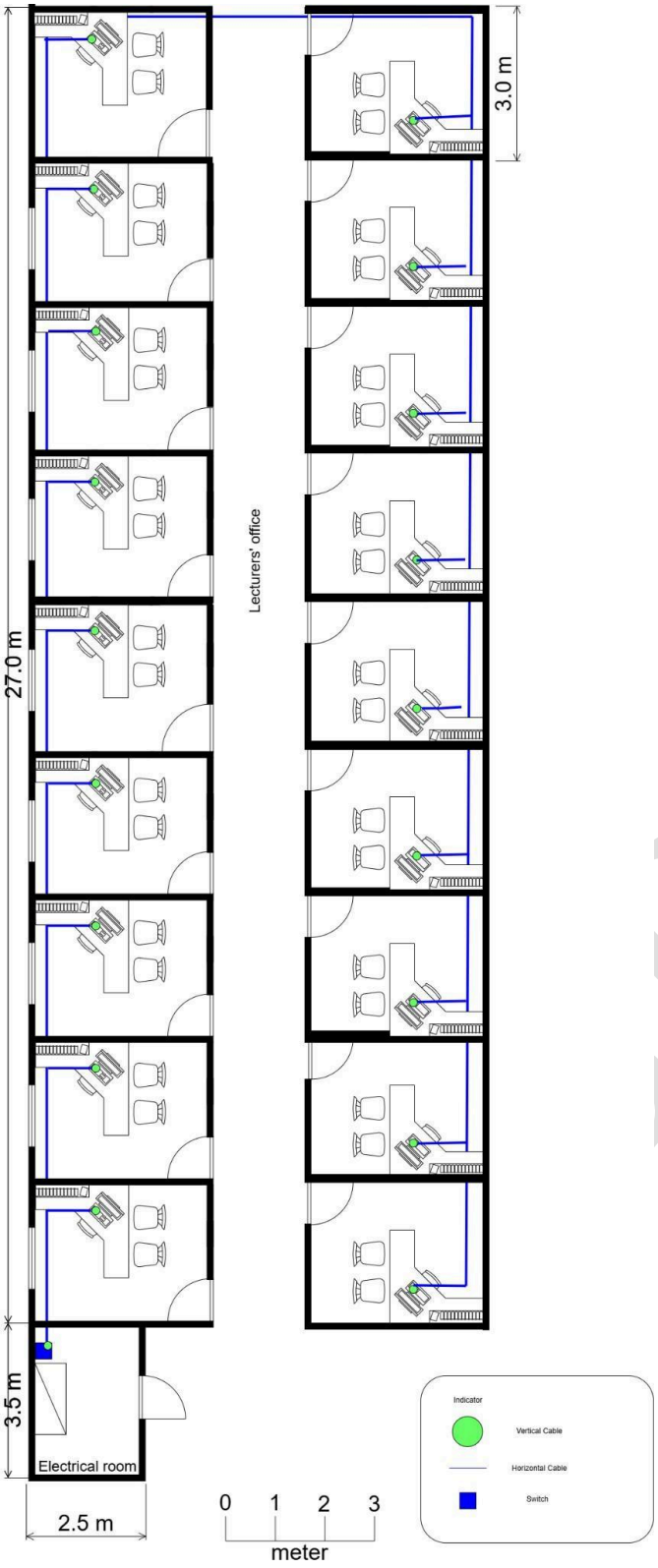


Figure 3.2.3 : Lecturer's Office & Electrical Room

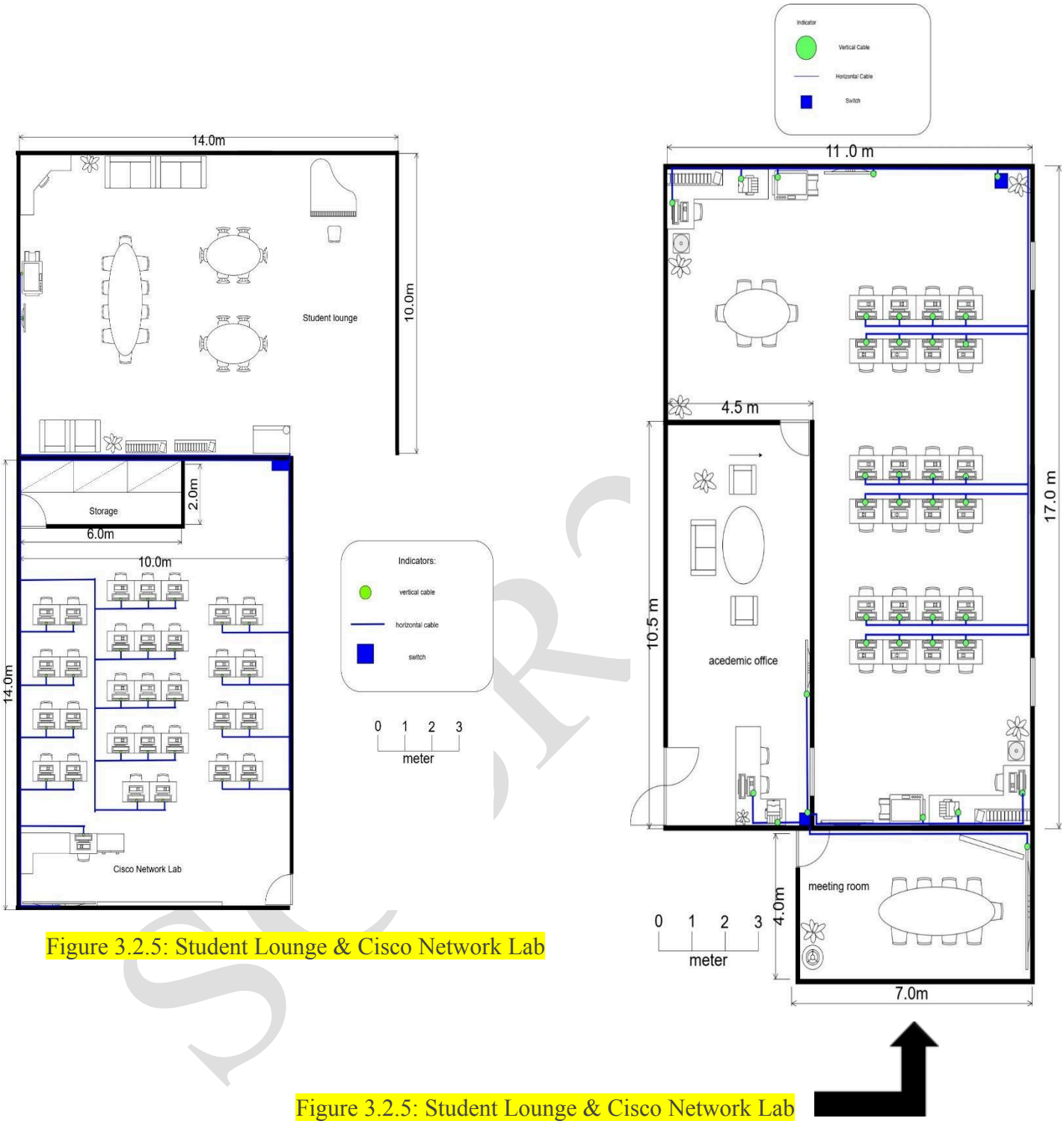
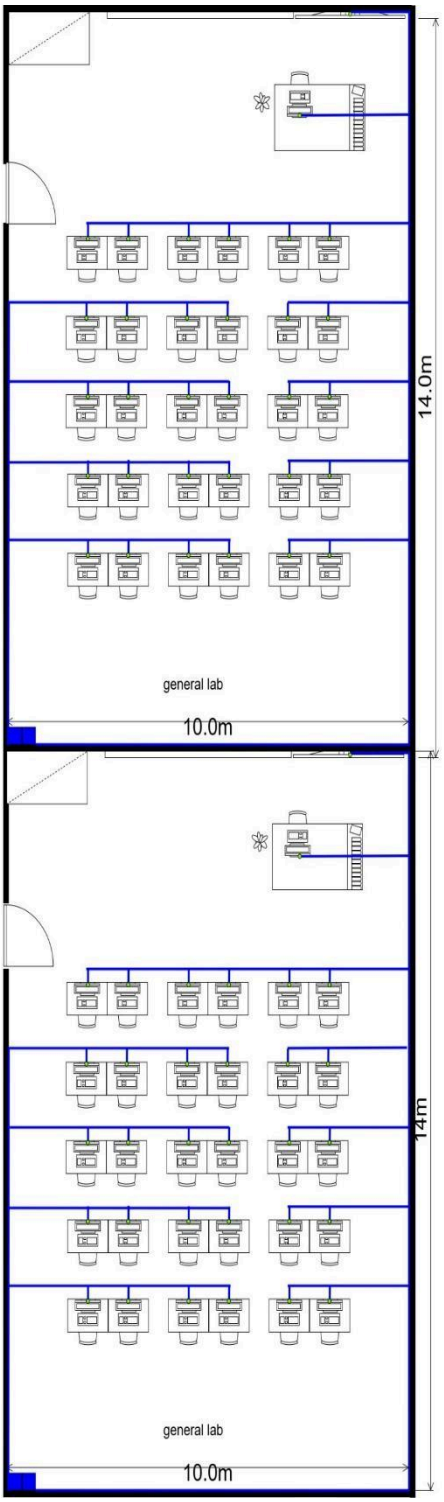
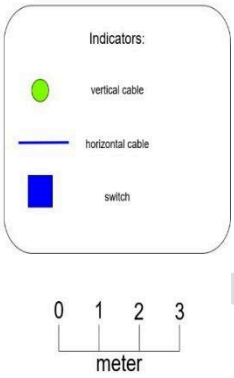
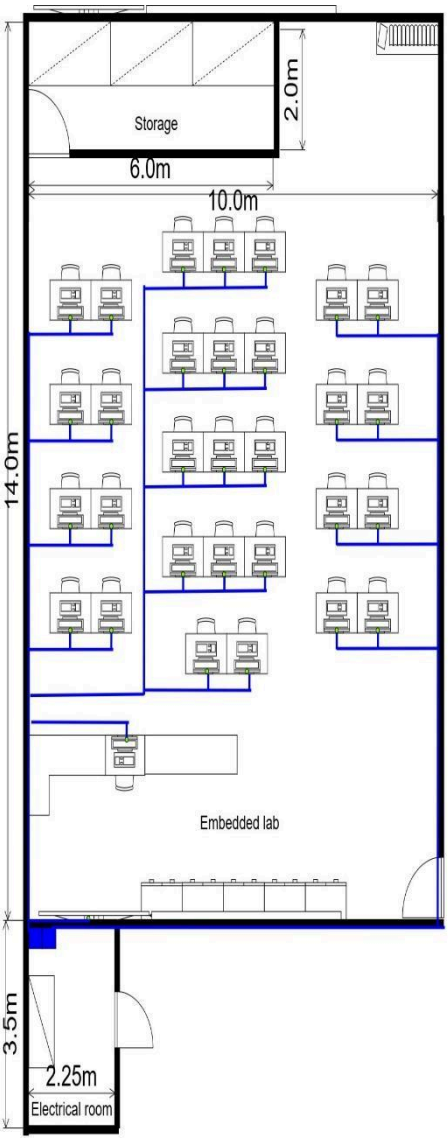


Figure 3.2.5: Student Lounge & Cisco Network Lab

Figure 3.2.5: Student Lounge & Cisco Network Lab







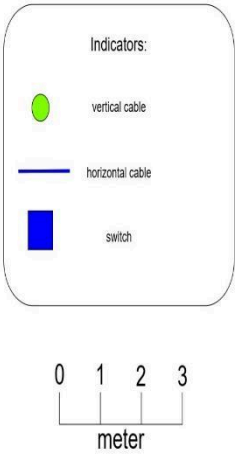


Figure 3.2.6 : Embedded Lab & Electrical room

Figure 3.2.7 : General Labs

Floor Diagram:

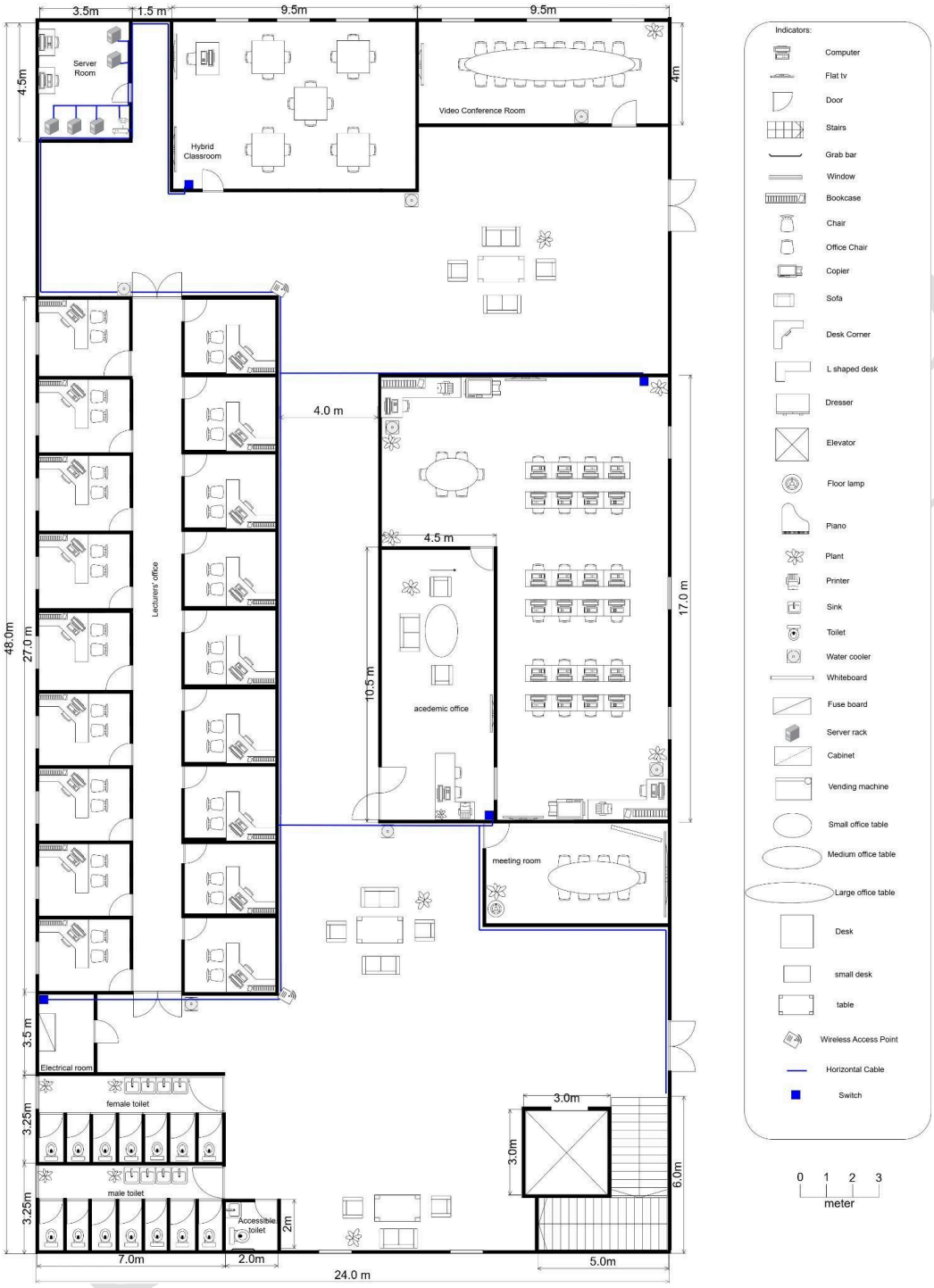


Figure 3.3.1 : Ground Floor

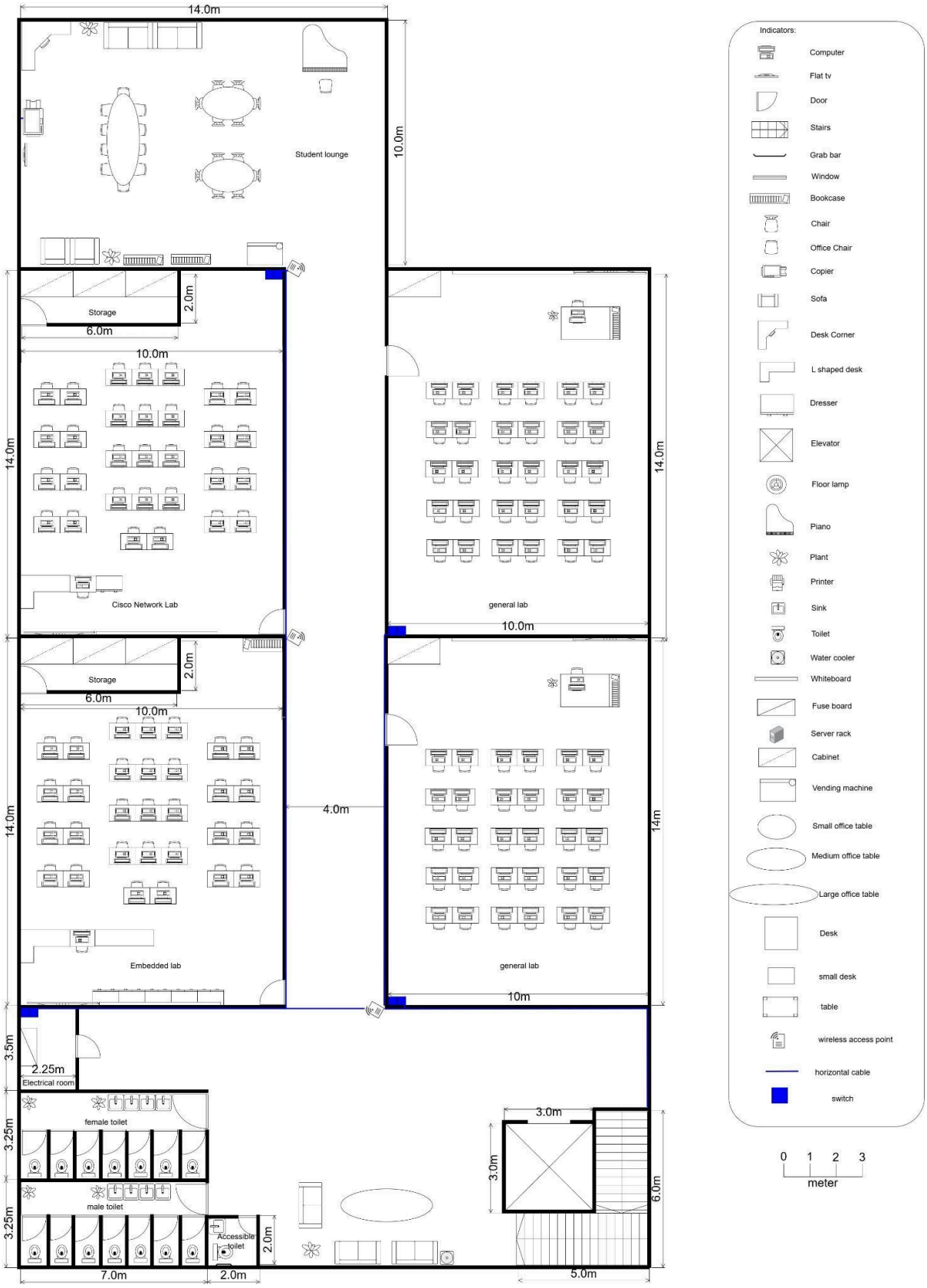

14.0m
Student lounge
10.0m
Storage
2.0m
6.0m
10.0m
14.0m
Cisco Network Lab
general lab
14.0m
10.0m
Storage
2.0m
6.0m
10.0m
4.0m
14.0m
Embedded lab
general lab
14m
10m
3.5m
2.25m
Electrical room
female toilet
3.25m
male toilet
3.25m
Accessible toilet
2.0m
7.0m
2.0m
3.0m
3.0m
6.0m
5.0m
Indicators:
Computer
Flat tv
Door
Stairs
Grab bar
Window
Bookcase
Chair
Office Chair
Copier
Sofa
Desk Corner
L shaped desk
Dresser
Elevator
Floor lamp
Piano
Plant
Printer
Sink
Toilet
Water cooler
Whiteboard
Fuse board
Server rack
Cabinet
Vending machine
Small office table
Medium office table
Large office table
Desk
small desk
table
wireless access point
horizontal cable
switch
0 1 2 3
meter

Figure 3.3.2 : First Floor

Number of connections, patch cords and switch ports needed:

- **Number of Switch Ports Needed:**

Area	Number of Switch Ports Needed
Hybrid Classroom	3
Video Conference Room	1
Academic Office	34
Meeting Room	1
Lecturer's Office	18
Cisco Network Lab	32
Embedded Lab	32
General Lab 1	32
General Lab 2	32
Student Lounge	2
Total	187

- **Number of Patch Cords Needed:**

Area	Number of Patch Cords Needed
Hybrid Classroom	3
Video Conference Room	1
Academic Office	34
Meeting Room	1
Lecturer's Office	18
Cisco Network Lab	32
Embedded Lab	32
General Lab 1	32
General Lab 2	32

Student Lounge	2
Total	187

- **Number of Connectors Needed:**

Area	**Number of Connectors Needed**
Server Room	14
Hybrid Classroom	6
Video Conference Room	2
Academic Office	68
Meeting Room	2
Lecturer's Office	36
Cisco Network Lab	64
Embedded Lab	64
General Lab 1	64
General Lab 2	64
Student Lounge	4
Total	374

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Identifying cable lengths And Prices:

- **Lengths and Prices Table:**

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Room				
Server Room	9.7	2.8	12.5	83.38
Hybrid Classroom	17.6	1.6	19.2	269.18
Video Conference Room	1.4	0.2	1.6	22.43
Academic Office	126.9	73.4	200.3	2808.21
Meeting Room	7.4	0.6	8	112.16
Lecturer's Office	79.2	22.6	101.8	1427.24
Ground Floor Room Total			343.4	4722.6
First Floor				
Room				
Cisco Network Lab	84.9	25	109.9	1540.80
Embedded Lab	81.2	25	106.2	1488.92
General Lab 1	92.25	25	117.25	1643.85
General Lab 2	92.25	25	117.25	1643.85
Student Lounge	16	0.6	16.6	232.73
First Floor Room Total			467.2	6550.15
Path			Length(m)	Price of cables used (RM)

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Total Path Length of Ground Floor	110.1	1543.60
Total Path Length of First Floor	69.6	975.79
Total (Ground Floor Rooms + Fi		13792.14 RM



In the previous section, we discussed how devices in the building are interconnected and how the network is distributed. To ensure that these connections are reliable, selecting the right type of cables is crucial. After discussing as a team, we decided to use a combination of CAT6 twisted-pair cables and fiber optic cables.

CAT6 twisted-pair cables will be used for the internal wiring within the building, such as in labs and lecture halls. These cables are ideal because they support high data transfer speeds of up to 10 Gbps. The thicker copper wires in CAT6 cables help reduce heat, making them capable of handling high power levels. This not only saves time during installation but also reduces costs. We opted for the Double Shielded Twisted Pair (STP) version of CAT6, as it is more durable and has a longer lifespan.

Fiber optic cables will be used to connect the routers in the server room and link the building to the Internet Service Provider (ISP). These cables transfer data using light, making them faster and more reliable. They can carry large amounts of data over long distances without any loss in signal quality. Fiber optic cables are essential for ensuring a stable and strong connection.

In conclusion, the building will require both CAT6 and fiber optic cables. This combination will provide fast, reliable connections within the building and ensure a strong link to the ISP.

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Meeting Minutes 1

Date/Time	7June 2025 10:00pm
Location	Online “google meet”
Agenda	1. The task. 2. The suggested idea. 3. Task division. 4. Working on the project.
Meeting MC	Garrah Thabit Mohammed

Attendance

Name	Time	Reason For Absence
Garrah Thabit Mohammed	10:00 pm	-
MOHAMMED ALSAKKAF	10:00 pm	-
LEE TUNG ERN	10:30 pm	-
Mohammed Abdulrahim	10:10 pm	

Minutes

No.	Items Discussed	Ideas/Suggestions and Person Giving It	Person in Charge & Date

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1.	Identifying Work Area	Garrah initially identified the work areas and discussed them with the other team members. Once everyone agreed, he mentioned that he would finalize the labels after the meeting.	Garrah
2.	Connections	discussed the design of the network distribution diagram for the floor plan, and volunteered to create the diagram later and update the connection design.	Garrah
		The group also discussed how to approach the floor plan connection design, deciding that two members would each handle the design for one floor. They would also calculate the length of cables and their costs.	The Team
		volunteered to design the network connections for the ground floor, which includes six work areas.	LEE TUNG
		Garrah volunteered to design the connection for the first floor, including 5 work areas.	Garrah

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3.	Google docs	A collaborative document was created, and a table was set up to track the required cable lengths and types for the project.	Mohammed Abdulrahim
4.	Meeting ended	Meeting ended at11:30 pm.	-

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